

2PROJ - Subject and Rendering

Ouvert le : lundi 8 décembre 2025, 09:00

À rendre : dimanche 31 mai 2026, 23:59



1 - Introduction

Your team has decided to revolutionize the conflict and conquest strategy game **Risk**. You have decided to start developing **SupKonQuest**¹, a modernized version, more suited to current video game market. The primary objective of this project is to provide a first stable and saleable version that meets the stated specifications, and which will be presented to investors with the aim of raising funds and bringing the game to market.

You are free to use any language and technology you want. **You are free to use any game engine** such as Unity or Unreal Engine. You can propose a game that can be played via an executable or on a Web browser. You are obviously not allowed to copy or adapt an existing code. **You are not allowed to use a no-code game-making App or an in-game map editor (in addition, do not overuse blueprint-type tools).**

Failure to comply with the above instructions or the delivery rules will result in a grade of 0.

2 - Presentation

The goal is to conquer the entire map, or at least conquer enough to force the opponents to surrender. You are free to choose any theme or era for your maps, as long as it remains ethical and appropriate. The game can be played by 2 (1 + AI) to 8 players, online and/or against AIs. You are designing a 2D version, but you can also opt for isometric 3D.

At the start, each player randomly owns camps on the map; each player has the same number of camps, and some are not owned by any player (neutral). Players use camps to produce different types of units to attack their opponents' camps. Players regularly earn money in proportion to the number of camps they own, which they can use to make purchases. When a player owns an entire region, he/she unlocks additional money and bonus units appear regularly. A player loses as soon as he/she no longer has any camps. The map features a geography with various obstacles (forests, mountains, walls, etc.) as well as water with islands that can only be accessed by boats available at special camps on the waterfront.

Here is a playable version, which differs from the project in several ways, but gives an idea of how the game works: <https://www.warzone.com>.

The version created will be similar in many ways to the custom Warcraft III Risk maps.



Here is also a game recording of the W3 version:

3 - Functional expression

3.1 - General and setup

You will implement the game as specified in the previous section. Graphic quality and sound environment, playability and game balancing/design will be essential.

You will provide at least 3 different types of maps with different configurations (the initial camp assignments remain random, without giving an entire region to a player from the start). At regular intervals, the player receives money as indicated above, and potentially bonus troops. All camps produce the same units, with the exception of ports which offer ships, and neutral camps which are initially difficult to capture and offer expensive but advanced units. Each camp must be guarded by at least one troop; if the unit dies, the camp is lost and captured by the executing unit (the camp becomes neutral in the event of mutual death). The camp building acts as a turret and inflicts damage on enemies.

The game ends when one player owns all camps (or at least all non-neutral camps). If a player leaves, his/her camps and units become neutral.

Neutral camps and units are passive: there is no production or mobility, but they will attack within a certain range.

3.2 - Units

Producing units takes time, and a queue can be created. There are 7 types of unit: **Infantry** (basic no-range unit), **Support** (improves attack speed, movement speed, or damage; the player must activate the boost in the unit spells, which is limited in time and has a cooldown period), **Heal** (the player must also activate the spell in the unit spells, with a cooldown period), **Range** (basic ranged unit), **Heavy** (high damage, high HP, but slow and medium range), **Anti-Armor** (weak except against Heavy), and **Motar** (ranged-area attacks, the farther the target is from the impact center, the less damage it takes). Each is more or less effective against a particular type. They have different characteristics: Price, HP, Damage, Hit Speed, Build Time, Type, Walk Speed and Range. Special neutral camps offer essentially the same types of units, but more powerful.

Ports allow players to build 3 types of ships: **Transports** (used only to move a limited number of troops across water), **Frigates** (equivalent to a Range unit but on water), and **Destroyers** (equivalent to a Heavy unit but on water and with area damage). If a Transport ship sinks, the units being transported are lost.

Units can be freely moved by the player on the map (within the limits of the environment); the player can select a group of units and move them to a location.

This type of game requires micromanagement of units, so it is essential for players to be able to reposition their units during combat (even flee) or create shortcuts to access their unit groups (and spells, for example). The game is therefore played mainly with the mouse, and all actions are reflected in real time.

3.3 - Opponents

It is not about being connected to the same network and hosting a game on a player's computer, but about enjoying a fully online game. Players can create an account to play with players from all over the world. It is possible to search for opponents via matchmaking.

3 levels of artificial intelligence are available, ranging from more or less strategic, more or less aggressive. They can act in a scattered manner, or seek to optimize their territory and therefore their income.

3.4 - Experience

To meet market needs, your game will be available in at least 3 languages.

A leaderboard is available to display essential information such as the number of camps or income.

3.5 - Bonus

3 bonuses are possible:

- The game is available in isometric 3D.
- A map editor is available; allowing players to create maps, design the entire environment and position the camps. Players can compete on custom maps.
- The online gaming experience is enhanced by a friends and messaging system, profile customization (image, game statistics, etc.), and an Elo system for finding opponents.

4 - Delivery

Your rendering will be a ZIP containing your source codes, additional files (sounds, images, etc.), a user documentation and a technical documentation explaining your choices and/or implementation choices/details.

The user documentation is a game manual, is intended for any user and explains how to play. A user must have all the information to launch the game and understand all its details, rules, modes, etc.

The technical documentation is intended for professionals in the field, and should describe in-depth the choices and implementations. It will contain at least the following elements:

- Justification of the choice of programming language and graphics library.
- Description of the data structures.
- AIs behavior, pathfinding and game mechanics.
- Game design choices (formulas, statistics, prices, etc.).
- Network implementation.
- etc.

Your rendering must contain an executable, an installer, or a link in order to launch the game easily for a normal user, on several architectures. Any rendering that requires heavy installation of additional software or advanced user manipulation will be considered as inadequate. A game that implements disparate features without creating a functional game will also not be considered a valid rendering.

5 - Grading scale

The project will be graded as follows, on a 575/500 scale:

- Features – 200 points (**a score lower than 80, leads to the non-validation of the project**)
 - General game engine – 44 points
 - Design (playability, graphics, sounds, music, animations, diversity, etc.): 18 points
 - Physics management: 12 points
 - Menus and settings: 4 points
 - Initial setup: 7 points
 - End of the game: 3 points
 - Gameplay – 30 points
 - 3 maps: 7 points
 - Capturing camps: 5 points
 - Defending camps: 7 points
 - Regions: 2 points
 - Income: 4 points
 - Unit generation: 5 points
 - Units and boats – 45 points
 - Recruitment and queue: 4 points
 - Statistics (HP, Damage, etc.): 9 points
 - Behavior (attacks, ranges, areas, types): 10 points
 - Activables (spells): 7 points
 - Control (selection, movement, pathfinding, macros, transport): 15 points
 - Multiplayer – 30 points
 - General playability: 20 points
 - Matchmaking: 10 points
 - 3 AIs: 36 points (12 points each)
 - Internationalization: 10 points
 - Leaderboard: 5 points
- Code quality – 200 points (**a score lower than 80, leads to the non-validation of the project**)
 - The points will be applied with the same scale as for **Features**
 - Efficiency, readability, maintainability, reliability, robustness, testability and portability will be evaluated here (ex: suitable data structures, no code duplication, no global variables, naming conventions, code structure, separation between data structures and graphical management algorithms, compliance with object-oriented principles if the object paradigm is used, etc.)
- Documentations – 100 points (**a score lower than 50, leads to the non-validation of the project**)
 - Technical documentation (70 points)
 - User documentation (30 points)
- Bonus – 75 points maximum
 - Isometric 3D (20 points maximum)
 - Map editor (25 points maximum)
 - Advanced online experience (30 points maximum)

¹ Although a school subject, the realizations are the exclusive property of the students. SUPINFO will not take ownership of the exploitation rights, possible incomes or any further action taken on this project. The only use SUPINFO will have is to evaluate the work carried out; or promotional use with the students' consent.

Ajouter un travail

Statut de remise

Groupe	Paris - 4
Statut des travaux remis	Rien n'a été déposé pour ce devoir
Statut de l'évaluation	Non évalué
Temps restant	60 jours 1 heure restants
Dernière modification	-

Commentaires[▶ Commentaires \(0\).](#)**Contactez-nous****Suivez-nous** [Contacter l'assistance du site](#)

Connecté sous le nom « Asma Ben-sliman » (Déconnexion)